Sales Performance Dashboard

A detailed project report on analyzing regional and product-wise sales performance using SQL for data extraction and Power BI for visualization and KPI monitoring.

Project Objectives

1

Data Preparation

Cleaning & transformation using SQL.

2

KPI Monitoring

Calculation and tracking of key performance indicators.

3

Visualization

Interactive dashboard design in Power BI.

4

Insights

Generation of actionable business insights.

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Dataset Overview

Over 25,000 sales records were analyzed, including:

- Order ID & Date
- Region & Product Category
- Customer Segment
- Sales Amount & Quantity
- Discount & Cost Price

Data was stored in relational tables and joined using SQL.

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SQL Data Cleaning & Transformation

Data Cleaning

Removed duplicates, handled NULL values, standardized date formats.

Calculated Profit

Created a new column for profit calculation.

Table Joins

Used INNER JOIN to combine Sales, Product, and Customer tables.

Aggregations

Applied SUM(), COUNT(), GROUP BY, ORDER BY, and HAVING for data summarization.

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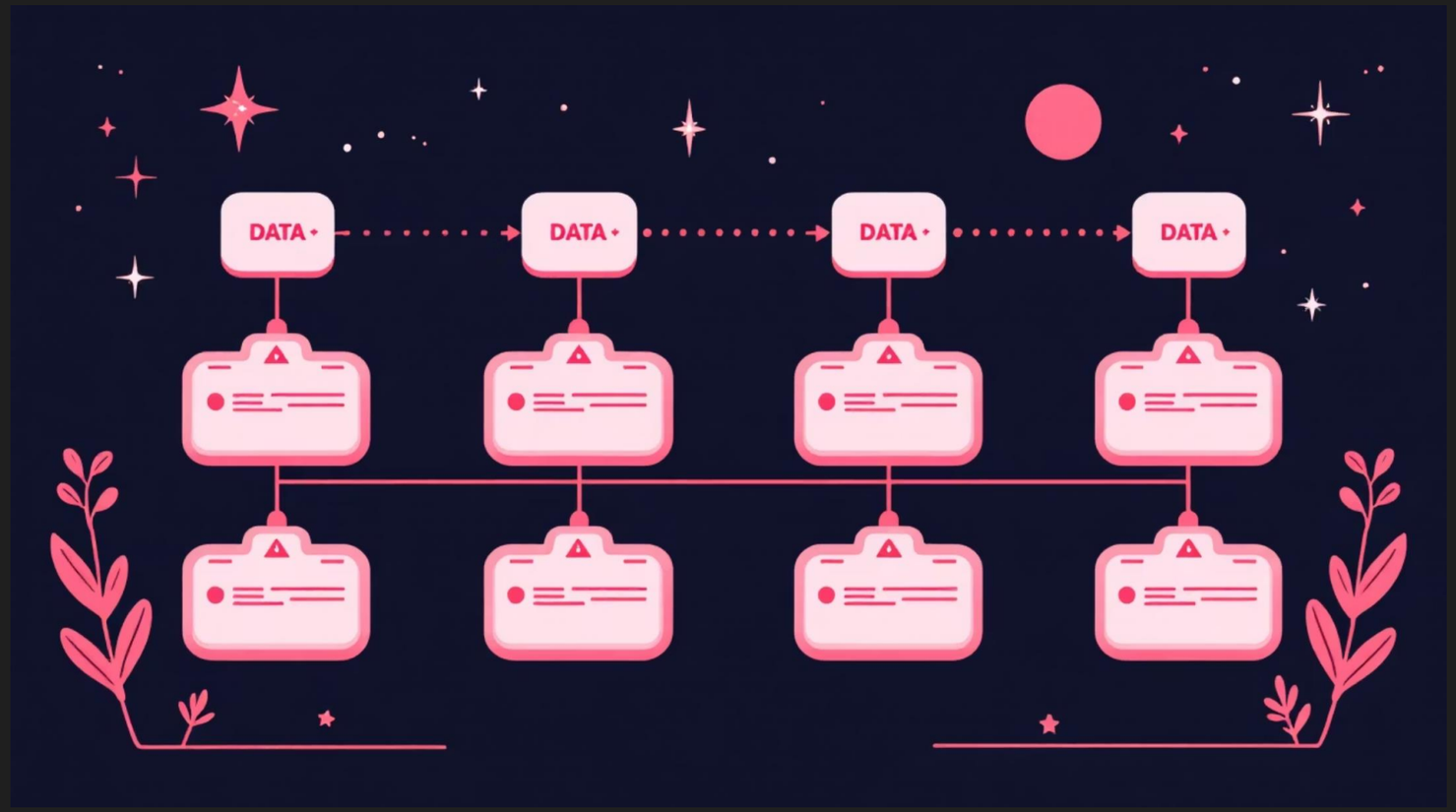
Power BI: Data Modeling

After SQL preprocessing, data was imported into Power BI. Key steps included:

- Creating relationships between tables.
- Implementing a Star Schema for optimal performance.
- Defining Primary & Foreign Keys.
- Creating a Date hierarchy (Year, Quarter, Month).



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Key Performance Indicators (KPIs) in Power BI

Important measures created using DAX:

Total Sales

SUM(Sales[Sales_Amount])

Total Orders

DISTINCTCOUNT(Sales[Order_ID])

Total Profit

SUMX(Sales, (Sales[Sales_Amount] - (Sales[Cost_Price] * Sales[Quantity])))

Profit Margin %

DIVIDE([Total Profit], [Total Sales], 0)

Sales Growth %

Month-over-month growth analysis.

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Dashboard Visuals

KPI Cards

- Total Sales
- Total Orders
- Total Profit
- Profit Margin %
- Average Order Value

Interactive Filters

- Region, Category, Date, Segment slicers

Charts

- Regional Sales (Bar Chart)
- Monthly Sales Trend (Line Chart)
- Category-wise Sales (Column Chart)
- Top 10 Products (Bar Chart)
- Customer Segment Distribution (Pie Chart)

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Key Insights Identified

Regional Decline

South region showed a 14% decline after the festive season.

Top Category

Technology category generated the highest revenue.

Customer Profit

Corporate customers contributed the highest profit.

Discount Impact

High discounts negatively impacted profit margins.

Pareto Principle

Top 20% products generated the majority of revenue.

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Business Impact

The dashboard empowered management to:

- Monitor sales performance in real-time.
- Identify low-performing regions.
- Optimize discount strategies.
- Improve product focus.
- Make data-driven strategic decisions.

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Conclusion & Tools Used

Project Success

Successfully transformed raw sales data into actionable insights, enabling effective performance tracking and strategic decision-making.

Tools & Technologies

- SQL: Data extraction & transformation
- Power BI: Visualization & dashboard creation
- DAX: KPI calculations
- Excel: Initial inspection

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